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Deterritorialization and reterritorialization of knowledge: immersive experience spaces in national libraries in China and the United Kingdom

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Abstract

Purpose – This paper uses Deleuze and Guattari's theory of deterritorialization to examine how immersive library experience spaces transform the forms, organization and circulation of knowledge and to propose strategies for balancing knowledge fluidity with systematic order.

Design/methodology/approach – A theory-driven multiple-case study compares the National Library of China and the British Library, drawing on public project documents, reports, media coverage and scholarly commentary. The analysis traces changes in carriers, organization, communication, production and access.

Findings – The findings show that immersive library spaces do not simply make knowledge more accessible; they alter the conditions under which knowledge is stabilized, connected, authorized and experienced.

Research limitations/implications – The study relies on public secondary materials and two national-library cases, so its transferability to smaller libraries requires adaptation.

Originality/value – The paper introduces the deterritorialization theory into the analysis of immersive library knowledge services and offers a five-dimensional framework for understanding knowledge transformation in digital library settings.

Keywords Libraries, Immersive experience spaces, Deterritorialization, Reterritorialization, Knowledge services, Deleuze and guattari, Case study

Paper type Research article

1. Introduction

Immersive experience spaces have become an important form of knowledge service in the digital transformation of libraries (Massis, 2015; Pope, 2018). The integration of digital technologies with immersive experience is not only changing how libraries display cultural materials, but also reshaping how knowledge is stored, organized, communicated and produced (Bekele *et al.*, 2018; Piccialli and Chianese, 2017). Libraries are moving beyond their conventional role as repositories for document storage and lending toward service spaces that combine knowledge experience, interactive co-creation and cultural communication. This shift is therefore not a superficial addition of new scenes and technologies. It represents a systemic reworking of library knowledge systems and service models.

Technologies such as virtual reality (VR), augmented reality (AR), knowledge graphs and data visualization loosen the established boundaries of knowledge carriers, organizational modes and communication pathways (Bizer *et al.*, 2023; Zeng, 2019). They enable a transition from a static, segmented and institutionally enclosed knowledge form to one that is mobile, relational and perceptible. In immersive settings, paper-based carriers can be digitized and placed into multiple flows; hierarchical classification can be supplemented by networked association; one-way communication from librarian to reader can become multi-node

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interaction; authority-led production can be opened to collaborative creation and passive retrieval can be transformed into embodied exploration.

Existing library and information science research has examined immersive spaces mainly in terms of user experience, service innovation, technological application and spatial operation. Less attention has been paid to what these spaces do to knowledge itself: how documentary materials are detached from inherited carriers and classifications, how they are reorganized through digital infrastructures and where the limits of such reorganization should be drawn. This paper addresses that problem by using Deleuze and Guattari's concepts of deterritorialization and reterritorialization as an analytical lens. It asks how immersive library experience spaces unsettle and recompose knowledge carriers, organizational structures, communicative relations, production roles and access practices. By doing so, the article shifts the discussion from the adoption of immersive technologies to the deeper question of how libraries maintain intelligible, trustworthy and public knowledge under conditions of digital fluidity.

2. Deterritorialization theory and its applicability

Deterritorialization and reterritorialization, developed in Deleuze and Guattari's work on signs, desire and social formations, provide a vocabulary for analysing how relatively stable arrangements are loosened, displaced and recomposed (Deleuze and Guattari, 1983, 1987). In this article, their value lies less in offering a general philosophical framework than in making visible a double movement within immersive library services. Digital imaging, VR/AR interfaces, semantic linking and interactive design unsettle inherited relations among documents, collections, classifications and readers. At the same time, these loosened elements have to be reassembled through metadata, interface design, curatorial interpretation and institutional governance if they are to remain intelligible, trustworthy and useable. Deterritorialization is therefore understood here not as simple dissolution, but as a process inseparable from reterritorialization.

A small but suggestive body of information and documentation scholarship has already shown how Deleuzian concepts can illuminate libraries, documents and systems of organization. Robinson and Maguire (2010) use the contrast between rhizomatic and arborescent structures to rethink information organization; Gerolami (2015) approaches the library as an assemblage with creative institutional potential and Faucher (2014) draws on Deleuzian thought to reconsider documentation as a more dynamic informational condition. This paper extends that line of work by asking how immersive library spaces destabilize and reorder knowledge carriers, organizational structures, communicative relations, production roles and access practices.

2.1 Deterritorialization and reterritorialization as a dialectical developmental logic

Deterritorialization and reterritorialization are the central concepts of the theoretical framework used in this article (Deleuze and Guattari, 1987). Deterritorialization refers to the process by which signs, knowledge, desires or other elements leave their original contexts, rules or coding systems, such as disciplinary classification or authoritative discourse, thereby releasing elements and space for new forms. Reterritorialization refers to the subsequent reconnection and aggregation of these loosened elements under new conditions, through which structures and meanings are rebuilt. The two processes are interdependent: they do not occur as two simply sequential stages, but operate as a continuous dialectical movement in which each reterritorialization contains the possibility of another deterritorialization.

From an epistemological perspective, knowledge deterritorialization is the process through which knowledge escapes the traditional constraints of physical carriers, disciplinary boundaries and institutional frameworks and becomes a set of digital units that can flow, recombine and be perceived. Knowledge reterritorialization is the process through which these

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mobile elements are re-aggregated in new technological environments, given a renewed order and made meaningful for use. Deterritorialization releases the elements required for reterritorialization; reterritorialization gives mobile knowledge an ordered structure. Their dialectical coordination forms a deeper motor of knowledge transformation (Figure 1).

2.2 Arborescent and rhizomatic structures: two logics of knowledge organization

Arborescent structure and rhizomatic structure are two spatial forms developed in Deleuzian thought and are useful here as two logics of knowledge organization and experience construction (Deleuze and Guattari, 1987; Robinson and Maguire, 2010). Their essential difference lies in the tension between fixity and fluidity in knowledge organization and spatial order. The arborescent structure is centralized and hierarchical. Traditional library knowledge organization and physical spatial layout have largely relied on this model: linear systems organize knowledge for orderly storage and precise retrieval.

The rhizomatic structure is decentralized and multi-connected. In immersive library experience spaces, it becomes a dominant logic of organization because it does not presuppose a single centre or hierarchy. Knowledge nodes can be connected or disconnected according to need, allowing knowledge to move beyond fixed forms and exist as a relational, extensible and mobile field.

2.3 Flow and becoming as internal dimensions of deterritorialization

In Deleuzian philosophy, flow and becoming are two internal dimensions of deterritorialization (Deleuze and Guattari, 1987, 1996). Flow refers to the state of movement that follows the loosening of original codes. Becoming refers to the creative process toward which this movement is directed: entities continually exceed their boundaries and transform toward heterogeneous forms without a final endpoint. Flow and becoming are inseparable. Flow is the dynamic form of becoming, while becoming is the value orientation of flow. Without flow, becoming cannot occur; without becoming, flow may decline into disordered drifting.

In library immersive experience spaces, the flow of knowledge, such as digitized carriers and networked organization, is valuable because it can support new forms of knowledge becoming, such as cross-disciplinary cognition and new relations between readers and texts. This relation shows that deterritorialization is not merely the destruction of boundaries; it has a constructive orientation.

2.4 Applicability and limitations of deterritorialization theory

Deterritorialization theory treats knowledge space as a dynamic and continuously evolving carrier. This makes it suitable for interpreting knowledge deterritorialization and reterritorialization in library immersive experience spaces. Its applicability can be summarized in three ways. First, it highlights the dynamic generativity of the knowledge

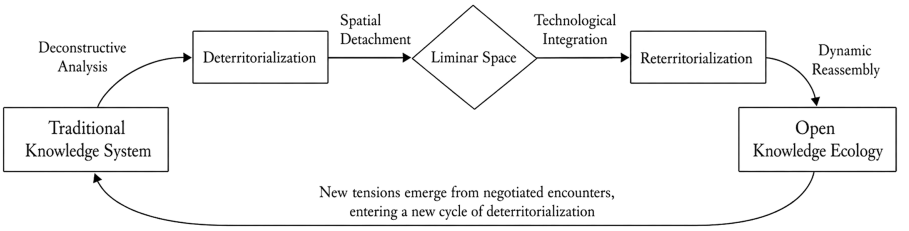


Figure 1. Dialectical relationship between knowledge deterritorialization and reterritorialization. Source: Authors' own work

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field by explaining how technological iteration and changing user needs drive knowledge beyond rigid boundaries. Second, it attends to the reconstruction of knowledge-power relations by analysing how deterritorialization disassembles the centralized and hierarchical knowledge monopoly of the traditional library, while reterritorialization clarifies new relations of production, circulation and sharing. Third, it affirms multiplicity and heterogeneity, offering a way to understand how libraries can remove knowledge barriers, respond to differentiated needs and promote cross-boundary knowledge integration.

The theory also has limitations for library practice. It privileges macro-level transformation and does not by itself explain all institutional constraints. Reterritorialization is not unlimited free reconstruction: public cultural institutions must retain publicness, service norms and operational efficiency. The rhizomatic structure is compatible with immersive knowledge organization, but excessive decentralization can weaken the stability and continuity required for knowledge inheritance. In addition, the theory is philosophically abstract and offers limited direct guidance for concrete tasks such as knowledge presentation, classification or service design, where classification infrastructures and naming practices have their own social and institutional consequences (Bowker and Star, 1999; Olson, 2002). It therefore needs to be combined with knowledge management, cognitive psychology and library service practice.

This study consequently adopts a dialectical perspective. It examines not only how technology deterritorializes established knowledge order, but also how new orders, meaning systems and service norms are reterritorialized so that deterritorialization does not proceed without adequate reconstruction.

3. Method

3.1 Methodological approach

The study uses a theory-driven case-study method. Its purpose is not to test hypotheses or make statistical claims, but to use the deterritorialization theory as an interpretive lens for explaining the transformation of knowledge forms in library immersive experience spaces. This approach is appropriate for exploratory and theory-building research, especially when the research question concerns how and why deeper mechanisms operate in complex settings. The research design draws on established multiple-case-study logic in organizational and information systems research, while adapting it to library and documentation studies (Eisenhardt, 1989; Walsham, 1995; Yin, 2018).

3.2 Case selection

The cases are immersive experience projects associated with the National Library of China and the British Library. They were selected according to three criteria. The first is typicality: both institutions hold world-class documentary heritage, represented by the Yongle Dadian and the Lindisfarne Gospels, and their immersive projects represent advanced digital-experience practice in libraries. The second is comparability: the two institutions are rooted in different cultural contexts, and their technological paths differ, with broad multi-terminal digital dissemination in the Chinese case and more site-based immersive experience in the British case. This contrast helps reveal multiple forms of deterritorialization and reterritorialization. The third is data accessibility: the two projects have relatively rich public documentation, media reporting and project descriptions suitable for qualitative analysis.

3.3 Data collection

Data were collected from public documents, including library websites, project pages, press releases and annual reports; media reports and scholarly commentaries in China and abroad and available technical specifications, exhibition descriptions, brochures and curator interviews. The data-collection period runs from December 2021, when the 5G panoramic VR project based on the Yongle Dadian was publicly reported, to March 2025; it also draws on

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documentation for the Yongle Dadian high-definition image database formally released in February 2023 (China Tourism News, 2021; National Library of China, 2023). For the British Library case, the data concern the Lindisfarne Gospels exhibition staged at the Laing Art Gallery, Newcastle, from 17 September to 3 December 2022, with the manuscript on loan from the British Library. They comprise exhibition descriptions, visitor statistics and records of school and community programmes (North East Museums, 2022), and fall within the same collection period. The study relies mainly on publicly available secondary materials and does not include internal decision-making records or micro-level user-interaction data. For an exploratory theoretical comparison across national contexts, however, multi-source public documentation remains a feasible and robust basis for analysis (Yin, 2018).

3.4 Analytical process

The analysis was guided by deterritorialization theory. First, an initial analytical framework was developed from core concepts such as deterritorialization, reterritorialization, arborescent structure and rhizome. These concepts supplied the analytical logic. The five dimensions of knowledge transformation, namely carriers, organization, communication, production and access, specified where that logic can be observed in library knowledge services; they draw on the processes of knowledge storage, organization, communication and production discussed in the Introduction (Bekele *et al.*, 2018; Piccialli and Chianese, 2017) and on themes recurring in an initial reading of the case materials. Each dimension is specified through four analytical elements, contrasted across traditional library and immersive settings and interpreted in terms of deterritorializing and reterritorializing effects (Tables 1–5). Second, the case materials were read closely to identify practices related to these five dimensions of knowledge transformation. Third, cross-case comparison was used to identify shared mechanisms and differences, and to refine the content of each dimension. The process followed an iterative dialogue between theory and data, and the author team discussed and checked the emerging interpretations against the theoretical framework (Eisenhardt, 1989; Walsham, 1995).

3.5 Trustworthiness and validity

Three strategies were used to strengthen trustworthiness. First, data-source triangulation combined project websites, press releases, scholarly commentary and technical descriptions to

Table 1. Transformation of knowledge carriers through deterritorialization and reterritorialization

Analytical element	Traditional library	Immersive experience space	Deterritorialization-reterritorialization effect
Carrier form	Tangible physical carriers, such as printed books and rare editions; difficult to disaggregate	Intangible digital carriers, including digital encoding; divisible and recombinable	Breaks dependence on physical media and establishes digital carrier forms
Spatial dependence	Tied to collection space; storage location is relatively fixed	Distributed storage; transferable across terminals and platforms	Reduces spatial constraint and enables efficient, ordered knowledge flow
Mode of use	On-site borrowing and consultation; cumbersome process and low reuse efficiency	Online access and flexible reuse; supports use across multiple scenarios	Simplifies service processes and increases knowledge-use efficiency
Preservation mode	Static storage vulnerable to physical deterioration	Dynamic cloud-based storage, long-term preservation and real-time updating	Moves beyond material deterioration and supports dynamic, sustainable preservation

Source(s): Authors' own work

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Table 2. Transformation of knowledge organization through deterritorialization and reterritorialization

Analytical element	Traditional library	Immersive experience space	Deterritorialization-reterritorialization effect
Organizational structure	Disciplinary hierarchy and linear arborescent structure; fixed paths	Networked association and rhizomatic structure; multidimensional knowledge links	Breaks hierarchical constraint and builds an open networked structure
Knowledge linking	Relies on fixed classification categories; cross-disciplinary integration is difficult	Semantic association activates implicit cross-disciplinary relations	Crosses disciplinary boundaries and strengthens integration
Retrieval and organization logic	Manual categories dominate; retrieval paths are fixed and inflexible	Intelligent association and proactive aggregation adapt to personalized needs	Improves retrieval logic and service precision
System openness	Relatively closed and rigid; categories are hard to expand dynamically	Open, dynamic and iterative; adapts to rapid knowledge change	Breaks closure and increases adaptive capacity

Source(s): Authors' own work

Table 3. Transformation of knowledge communication through deterritorialization and reterritorialization

Analytical element	Traditional library	Immersive experience space	Deterritorialization-reterritorialization effect
Communication mode	One-way linear dissemination led by the library; readers receive passively	Multi-node interactive communication with multiple actors and two-way exchange	Breaks centralized dissemination and enables participatory communication
Temporal-spatial constraints	Limited by opening hours and physical space	Integration of virtual and physical space; breaks venue and schedule limits	Extends the scope and accessibility of knowledge dissemination
Subject relations	Single dominant communicator and concentrated discursive power; low reader participation	Plural communicators; readers annotate, comment and co-create content	Reconfigures subject relations and strengthens reader agency
Feedback mechanism	Limited real-time feedback and relatively closed dissemination chain	Real-time interactive feedback; communication and optimization proceed together	Builds a feedback loop and improves communication effectiveness

Source(s): Authors' own work

cross-check factual claims. Second, researcher triangulation was pursued through joint review and discussion within the author team. Third, theory-guided analysis ensured that the data were interpreted in relation to the conceptual vocabulary of deterritorialization, while still allowing the empirical material to refine the framework (Yin, 2018).

4. Internal mechanisms of knowledge deterritorialization and reterritorialization in immersive experience spaces

In library immersive experience spaces, the internal mechanism of knowledge transformation can be understood as a twofold movement of loosening and recomposition. Digitization, semantic association, VR/AR interfaces and interactive design detach knowledge from some

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Table 4. Transformation of knowledge production through deterritorialization and reterritorialization

Analytical element	Traditional library	Immersive experience space	Deterritorialization-reterritorialization effect
Production subject	Single or concentrated, led by professional institutions and academic authorities	Plural, with professionals and ordinary users participating together	Breaks subject monopoly and forms a multi-party co-creation pattern
Production mode	Closed process with limited user participation	Open co-creation supporting editing, annotation and content generation	Opens production and activates user participation
Content generation	Top-down unified release with relatively fixed sources	Multiple sources and scenarios generate richer content	Broadens content sources and enriches knowledge supply
Quality control	Professional review and stable controlled sources	Plural production combined with normative management	Balances openness and order while protecting output quality

Source(s): Authors' own work

Table 5. Transformation of knowledge access through deterritorialization and reterritorialization

Analytical element	Traditional library	Immersive experience space	Deterritorialization-reterritorialization effect
Access mode	Active retrieval and on-site consultation; relatively fixed procedures	Scenario-based immersive experience; natural exploration and discovery	Introduces bodily participation and diversifies access forms
Presentation form	Textual and paper-based; flat presentation	Multimedia, contextualized and three-dimensional presentation	Moves beyond single presentation and enables multidimensional perception
Participation mode	Mainly consultation and indirect participation	Multi-sensory, bodily and interactive access	Shifts from indirect decoding to direct perception
Cognitive experience	Primarily relies on textual understanding	Situational immersion and multi-sensory coordination	Moves cognition from abstraction toward embodied knowing

Source(s): Authors' own work

of its inherited anchors: the physical document, the fixed shelf, the disciplinary class, the expert-centred voice and the retrieval terminal. Yet these elements do not remain in a state of free circulation. They are reassembled through metadata, platforms, curatorial narratives, interface pathways and governance rules into forms that users can navigate and institutions can sustain.

This movement is not linear. It unfolds across five interrelated dimensions: carriers, organization, communication, production and access. These dimensions are analytically separable, but in practice they reinforce one another. A digitized rare book, for example, is not only a new carrier; it can also enable semantic linking, public interpretation, user participation and embodied encounter. Culture is therefore no longer only an object to be preserved and explained by institutions; it becomes a participatory field in which readers help construct relations among documents, meanings and contexts. This shift from passive reception to participatory construction is also visible in cultural-heritage crowdsourcing and participatory GLAM practice (Oomen and Aroyo, 2011; Ridge, 2014) (Figure 2).

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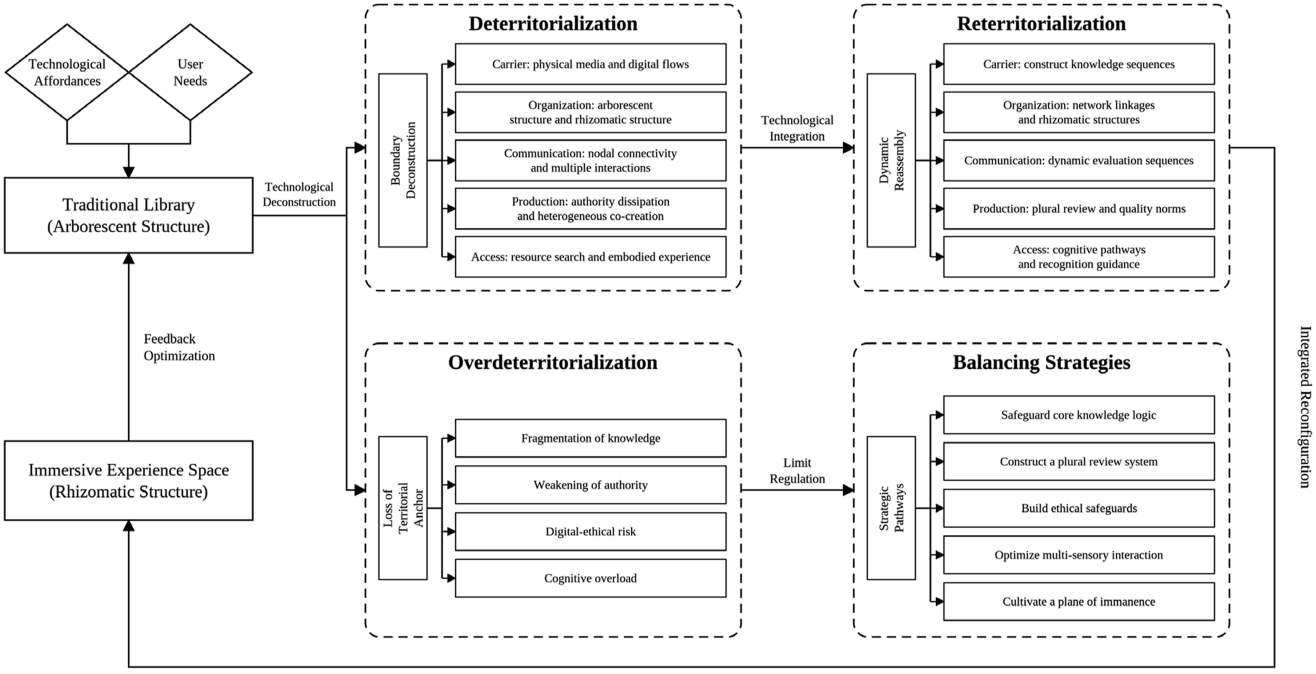


Figure 2. Internal mechanism of knowledge deterritorialization and reterritorialization. Source: Authors' own work

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4.1 Knowledge carriers: from physical fixity to digital flow

Traditional library knowledge is carried primarily by physical documents. Because it depends on material carriers and collection space, it tends toward fixity. Under the influence of digital technologies, knowledge can break through the boundaries of physical media and space, becoming digital units that can flow, be decomposed and be recombined. This deterritorializes knowledge carriers and allows digital surrogates of rare books and manuscripts to exist independently of their physical originals (Piccialli and Chianese, 2017; Zeng, 2019).

On this basis, libraries reterritorialize dispersed digital knowledge units through semantic indexing, data governance and resource-management systems. These practices maintain internal relations and overall order, and shift the library's role from guardian of physical knowledge to constructor of digital order. The dialectical coordination of deterritorialization and reterritorialization thus supports the transition from physical documents to digital knowledge space and lays the foundation for immersive knowledge services (Bizer *et al.*, 2023; Zeng, 2019).

4.2 Knowledge organization: from hierarchical classification to networked association

Traditional libraries organize knowledge mainly through disciplinary classification and hierarchical belonging, forming a relatively closed and linear arborescent structure. This model ensures systematic management but restricts knowledge flow, makes cross-domain connection difficult and often leaves implicit relations inactive (Bowker and Star, 1999; Olson, 2002). In immersive experience spaces, technology enables knowledge units to detach from a single classification system and become mobile nodes. Semantic association further loosens knowledge boundaries and creates conditions for recombination.

Libraries can then use knowledge graphs to sort and link deterritorialized nodes, constructing a decentralized and multi-node rhizomatic structure (Bizer *et al.*, 2023; Zeng, 2019). Reterritorialization is therefore not a return to the old hierarchy; it is the building of a new order that is both open and navigable. The interaction between deterritorialization and reterritorialization moves knowledge organization from a closed hierarchy toward an open network ecology and activates the generative potential of hidden relations.

4.3 Knowledge communication: from one-way linearity to multi-node interaction

Traditional library knowledge communication is often organized as one-way dissemination. Readers receive knowledge passively, while discursive power is concentrated in the library. Physical space and opening hours add further limits, and the absence of real-time interaction reduces the efficiency and reach of communication. In immersive experience spaces, real-time connectivity and VR break the centralized communication pattern. They deterritorialize knowledge communication by reducing spatial and temporal constraints, transforming readers into participants and supporting annotation, comments and content co-creation (Massis, 2015; Oomen and Aroyo, 2011).

Reterritorialization occurs when libraries integrate interactive feedback and regulate communication order. The aim is not to suppress participation, but to maintain norms while preserving reader agency. This dialectic moves knowledge communication from one-way linear dissemination to multi-directional interaction and creates a communicative basis for services that can be interacted with, evaluated and iteratively improved.

4.4 Knowledge production: from authority-led production to plural co-creation

Traditional knowledge production is led by professional institutions and academic authorities. Production subjects are concentrated, processes are closed and ordinary users mainly receive rather than generate, review or publish knowledge. In immersive experience spaces, mechanisms such as crowdsourced editing, user-generated content and collaborative co-creation deterritorialize knowledge production by breaking the monopoly of a single subject.

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Professionals and users can participate together in generating, annotating and commenting on knowledge, making production more open (Holley, 2010; Oomen and Aroyo, 2011; Ridge, 2014).

Libraries then reterritorialize plural knowledge production through content review, quality norms and copyright management. These mechanisms do not erase openness; they give it a framework within which knowledge remains credible and useable. The dialectical movement from deterritorialization to reterritorialization shifts knowledge production from authority-led supply to open co-creation and supports a plural and sustainable knowledge-supply system.

4.5 Knowledge access: from retrieval and consultation to embodied experience

Traditional library knowledge access relies on active retrieval and on-site consultation. Readers often need retrieval skills and disciplinary background knowledge, while understanding depends on the indirect decoding of textual symbols. The process is primarily cognitive and lacks bodily sensory participation. In immersive experience spaces, VR, AR and scenario reconstruction deterritorialize knowledge access by breaking the reader's existing cognitive limits. Readers become participants in a scene and interact directly with knowledge through the body and the senses. Abstract knowledge is transformed into contextualized content that can be perceived (Dede, 2009; Makransky and Petersen, 2021).

Reterritorialization occurs through scene design and cognitive-pathway guidance. The library's role shifts from resource provider to experience designer. The coordination of deterritorialization and reterritorialization renews access modes and creates conditions for the regular provision of immersive knowledge services.

5. The supporting value of immersive digital technologies in practice

Knowledge deterritorialization and reterritorialization cannot be realized without immersive digital technologies. In library immersive experience spaces, these technologies are not generic tools. They are mediating forces embedded in the dialectic of deterritorialization and reterritorialization. They break the rigid constraints of traditional knowledge systems and enable boundary-crossing recombination, while their systematic technical logic also guides the ordered reassembly of knowledge elements. The two cases considered here, the National Library of China's immersive work around the Yongle Dadian and the British Library's immersive digital experience around the Lindisfarne Gospels, illustrate how this mediating role operates in practice (China Tourism News, 2021; National Library of China, 2023; North East Museums, 2022).

5.1 Knowledge deterritorialization: technological deconstruction of traditional boundaries

The fixity of traditional knowledge systems is visible in the constraints of physical carriers, disciplinary separation and singular access scenarios. Immersive digital technologies loosen these constraints and make systematic deterritorialization possible.

5.1.1 Carrier deterritorialization: breaking physical dependence and enabling knowledge flow. The National Library of China's VR project based on the Yongle Dadian used 5G and VR techniques, 8 K panoramic video and three-dimensional animation to transform textual and historical materials into an immersive reconstruction of the rare work (China Tourism News, 2021). In parallel, the high-definition image database of the Yongle Dadian makes the National Library of China's surviving holdings more widely available through digitized images, text and knowledge annotations (National Library of China, 2023; Research Center for Digital Humanities of PKU, 2023). The British Library's Lindisfarne Gospels were displayed at the Laing Art Gallery in an exhibition that opened with an immersive digital experience and placed a rare manuscript within a wider interpretive environment (North East Museums, 2022). Materials that were once restricted by conservation conditions and could be displayed

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only in limited library contexts were transformed into digital knowledge units capable of repeated use and wider public communication.

5.1.2 Disciplinary deterritorialization: crossing classification barriers and enabling integration. The Yongle Dadian VR project used three-dimensional visualization and semantic association to transform textual, visual and historical materials into interactive scenes. It linked literature, bibliography, history and material culture, turning dispersed knowledge into a network of relations (China Tourism News, 2021; Research Center for Digital Humanities of PKU, 2023). The Lindisfarne Gospels exhibition connected religious history, art history, manuscript studies and regional cultural identity by placing the manuscript alongside other early medieval objects and new interpretive media (North East Museums, 2022). In both cases, multimodal technologies supported the disciplinary deterritorialization of ancient-book knowledge in different cultural contexts.

5.1.3 Scenario deterritorialization: breaking spatial limits and widening access channels. The National Library of China adapted digital knowledge units from the Yongle Dadian to VR display and online database environments, allowing users to encounter rare-book knowledge beyond the ordinary reading-room setting (China Tourism News, 2021; National Library of China, 2023). The British Library's Lindisfarne Gospels experience placed manuscript interpretation in an on-site immersive zone and extended public engagement through schools and community programmes (North East Museums, 2022). The two institutions followed different paths, device-based and online dissemination in the Chinese case and environmental immersion in the British case, yet both broke the spatial limits of knowledge access.

5.2 Knowledge reterritorialization: technological construction of order

The value of immersive digital technologies lies not only in deterritorialization but also in reterritorialization. Through systematic technical design, they guide loosened knowledge elements into ordered recombination and prevent fragmentation or disorder. This is essential for maintaining the professional and systematic qualities of library knowledge services.

5.2.1 Carrier reterritorialization: building digital order and protecting knowledge security. On the basis of digitized reproduction, public documentation of the Yongle Dadian work emphasizes open service, knowledge-oriented processing and the continuing need for copyright protection and data-asset management in ancient-book digitization (National Center for Preservation and Conservation of Ancient Books, 2025; Research Center for Digital Humanities of PKU, 2023). The British Library case likewise shows that rare manuscript display depends on exhibition governance, loan conditions and institutional curatorship as much as on visual mediation (North East Museums, 2022). These technical and institutional measures enable resources to flow while reducing the risks of disorderly circulation.

5.2.2 Disciplinary reterritorialization: integrating relations and building logical systems. The National Library of China case used knowledge annotation and structured processing to connect digitized images, texts and contextual information, building a network that supports more systematic access to the Yongle Dadian (National Library of China, 2023; Research Center for Digital Humanities of PKU, 2023). The British Library's immersive experience similarly integrated religious, artistic, historical and regional meanings through curatorial interpretation and exhibition design rather than through a single linear narrative (North East Museums, 2022). Both cases show how reterritorialization can transform fragmented contact with knowledge into more systematic cognition.

5.2.3 Scenario reterritorialization: optimizing experience design and clarifying cognitive goals. In the Yongle Dadian VR experience, temporal-spatial reconstruction provides a panoramic view, while micro-narrative restores textual and material details (China Tourism News, 2021). In the Lindisfarne Gospels experience, scenographic narrative and interactive guidance organize cross-disciplinary knowledge into a coherent sequence of manuscript interpretation, historical background and artistic appreciation (North East Museums, 2022).

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Different design languages are used, but both cases align immersive experience with knowledge-cognition goals and avoid cognitive deviation caused by excessive scenic richness.

5.3 Cross-case comparison and implications

5.3.1 Practical effects. The two cases are not identical institutional projects, but contrasting configurations through which rare documentary heritage is digitized, mediated and made experiential. The National Library of China's work on the Yongle Dadian combines VR presentation with a broader digital-resource infrastructure. The high-definition image database first released 40 volumes and 75 juan of the National Library of China's holdings, including high-resolution images, textual materials and sample knowledge annotations for later knowledge-system construction and intelligent use (National Library of China, 2023; Research Center for Digital Humanities of PKU, 2023). At the wider service-system level, the Chinese Ancient Books Resources Database had released 106,000 ancient-book items by the end of January 2025 and received more than 350 million visits in the previous three years under a no-registration access model (National Center for Preservation and Conservation of Ancient Books, 2025).

The Lindisfarne Gospels case follows a different path. Centred on a British Library manuscript and hosted at the Laing Art Gallery, the exhibition used immersive interpretation, object display and public programming to connect a rare manuscript with regional history, religious culture and community learning. It attracted 56,784 gallery visits over an 11-week period, three times the attendance of recent exhibitions at the venue; first-time visitors accounted for 23%. The accompanying schools programme delivered more than 550 workshops and generated over 25,000 engagements, and artist-led workshops in libraries and community settings added more than 1,500 community engagements (North East Museums, 2022).

5.3.2 Similarities, differences and lessons. The comparison points to two different routes into immersive knowledge service. The Chinese case emphasizes scale, openness and multi-terminal dissemination. Its primary contribution lies in moving rare-book knowledge from restricted physical access into a publicly available digital infrastructure. The British case emphasizes situated interpretation and community engagement. Its contribution lies in turning a manuscript encounter into a spatial, affective and educational experience.

Despite these differences, both cases show that immersive library services are not merely extensions of exhibition design or digital access. They reorganize the conditions under which knowledge is stabilized, connected, authorized and experienced. First, immersive technologies loosen the dependence of rare documentary heritage on the physical reading room or exhibition case. Second, institutions recompose the resulting knowledge elements through databases, metadata, curatorial pathways, interpretive design and public-service norms. Third, the two cases suggest that successful immersive knowledge services require a balance between openness and order: knowledge must be made mobile enough to circulate, but structured enough to remain meaningful, credible and reusable.

For libraries, the implication is that immersive experience spaces should not be evaluated only by visitor numbers, visual novelty or technological sophistication. Their deeper value lies in whether they create sustainable relations among documents, users, institutional expertise and public culture. The comparative analysis therefore supports the central argument of this paper: immersive digital technologies matter most when they mediate the double movement of knowledge deterritorialization and reterritorialization.

6. Limits of knowledge deterritorialization and balancing strategies

Deterritorialization breaks and reconstructs existing boundaries; it does not abolish boundaries altogether. In library immersive experience spaces, knowledge deterritorialization cannot be understood as unrestricted free deconstruction. Excessive deterritorialization detached from

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scientific governance and humanistic constraint may generate four kinds of risks: loss of systematicity, including fragmentation and breaks in meaning; weakening of authority, including information distortion and reduced trust; digital-ethical risk, including privacy leakage and algorithmic bias and cognitive overload, including attention dispersion and immersive disorientation (Bowker and Star, 1999; Makransky and Petersen, 2021; Mittelstadt *et al.*, 2016). These risks do not correspond one-to-one with the five dimensions of deterritorialization. They cut across dimensions. For example, knowledge fragmentation may arise from both carrier deterritorialization and organizational deterritorialization.

Clarifying the appropriate boundaries of knowledge deterritorialization and building feasible balancing strategies are therefore preconditions for a healthy cycle of deterritorialization and reterritorialization. These limits are not external denials of deterritorialization; they are expressions of the internal tension between deterritorialization and reterritorialization. The first four strategies below correspond to the four risks identified above. The fifth uses the Deleuzian plane of immanence to suggest a more fundamental path beyond external regulation.

6.1 Systematic limits: protecting core knowledge logic

Knowledge deterritorialization does not reject all order. It requires order to exist in a more open form, which is precisely the task of reterritorialization. The precondition of deterritorialization is therefore the preservation of core logics and basic relations within a knowledge system. Otherwise, excessive deterritorialization may lead to broken meaning and loose structure, especially where classification and metadata choices determine what can be found, connected and trusted (Bowker and Star, 1999; Olson, 2002).

Libraries can build a knowledge-guarding mechanism for immersive experience spaces in three ways. First, interdisciplinary teams of librarians, subject experts and cultural scholars should identify core knowledge boundaries, including historical sequences, key facts, literary backgrounds and textual meanings. Second, knowledge annotation systems should mark core knowledge nodes and protect them through read-only settings where appropriate, strengthening authority and recognizability. Third, dynamic updating mechanisms should recalibrate core knowledge boundaries in response to scholarly development and user feedback. Such practices show why reterritorialization is a necessary constraint on deterritorialization: without provisional order, knowledge can slide into meaningless fragments.

6.2 Authority limits: constructing a plural review system

Deterritorialization breaks the monopoly of a single authority, not authority as such. The key is to construct a modern knowledge-authority system characterized by plural participation and collaborative constraint. A plural review system is the concrete form of this new order in the reterritorialization process. If editing rights are fully opened without regulation, knowledge authenticity may decline and platform trust may be damaged; this tension is well documented in cultural-heritage crowdsourcing, where participation has to be balanced with quality assurance and institutional accountability (Holley, 2010; Oomen and Aroyo, 2011; Ridge, 2014).

Libraries should therefore establish layered review mechanisms. Factual knowledge, such as biographical and geographical data, should be examined by expert groups through screening, cross-verification and transparent review results. Interpretive and opinion-based content can be opened to user creation if authorship, sources and review status are clearly labelled. Provenance-aware systems, including but not limited to blockchain-based recordkeeping experiments, may record revisions, review points and modification histories, improving traceability and credibility (Lemieux, 2016). In essence, a plural review system is a decentralized form of reterritorialization: it preserves openness without creating an authority vacuum.

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6.3 Digital-ethical limits: building ethical safeguards

Technology-driven knowledge deterritorialization contains digital-ethical risks. Algorithmically shaped knowledge relations can create implicit cognitive discipline and intensify structural inequalities in access. Digital simulations and excessive deterritorialization may also weaken cultural authenticity and reduce the distinctive value of original knowledge systems. Privacy leakage and algorithmic bias can directly harm user rights and run counter to the public mission of libraries (Floridi *et al.*, 2018; Mittelstadt *et al.*, 2016; Nissenbaum, 2010).

A multi-level ethical safeguard system is therefore required. Privacy protection should anonymize behavioural data such as eye movement and interaction paths, and limit collection to data necessary for knowledge services. Algorithmic fairness and transparency should be maintained through regular audits of recommendation algorithms, monitoring for bias and interest lock-in. Content-ethics review should establish norms for virtual scenes and knowledge interpretation, clarify prohibited distortions or vulgarizations and respect cultural diversity and authenticity. Review teams including librarians, ethicists and cultural experts can conduct ex ante assessment and ongoing supervision. These safeguards are part of reterritorialization because they protect publicness and cultural authenticity within open knowledge flows (Floridi *et al.*, 2018).

6.4 Cognitive-load limits: optimizing multi-sensory interaction

Immersive technologies enhance knowledge perception through multimodal presentation, but excessive sensory stimulation can increase cognitive load, disperse attention, weaken knowledge internalization and produce experience fatigue. Balancing experience quality and cognitive efficiency requires layered cognitive intervention (Makransky and Petersen, 2021; Mayer and Moreno, 2003; Sweller, 1988).

At the physiological level, lightweight eye-tracking tools can monitor fatigue indicators such as pupil change and blink frequency, allowing the system to adjust brightness, dynamic speed or rest prompts when dizziness or attention decline appears. At the cognitive level, entry-level, intermediate and professional templates can adapt knowledge density and complexity to user background and interaction data. At the affective level, emotional-state recognition can trigger explanations or simplified scenes when users show frustration or difficulty. Cognitive-load control is a form of balance between deterritorialization and reterritorialization at the individual level because it prevents sensory overload from becoming loss of meaning (Dede, 2009; Makransky and Petersen, 2021).

These technologies can themselves become reterritorializing means that convert reader behaviour into new control records. Their use should therefore be transparent, authorized by users and accompanied by opt-out options. Libraries should follow a principle of minimal intervention and prioritize user-controlled adjustment.

6.5 Plane of immanence: an ethics of deterritorialization beyond external regulation

The first four strategies rely mainly on external regulation. Deleuzian philosophy also points to a more fundamental immanent path (Deleuze and Guattari, 1996). Genuine balance is not achieved solely by external authority setting limits, but by constructing a plane of immanence in which knowledge elements can form provisional consensus through the circulation of deterritorialization and reterritorialization. The library's role would then shift from external regulator to designer of a plane: it would not impose hierarchy and boundaries from above, but provide possibilities for multimodal connection so that users and knowledge can coordinate dynamically within flow.

Cognitive-load regulation offers an example. Automatic system downgrading may ease sensory overload, but a better approach is to allow users to define their own lines of flight, such as choosing sensory density, information complexity or interaction pace. This returns choice to users and moves the balance between deterritorialization and reterritorialization

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from external regulation toward immanent becoming. It respects individual cognitive autonomy and resonates more closely with Deleuzian concepts of becoming and desire production.

7. Conclusion

The deterritorialization and reterritorialization of knowledge in library immersive experience spaces reveal the internal logic through which immersive technologies break established knowledge structures and reconstruct cognitive order. Knowledge deterritorialization releases the flow and interactivity of knowledge by loosening rigid forms. Knowledge reterritorialization rebuilds order and protects core knowledge value. Their dialectical coordination expresses the integration of digital technology and humanistic value and forms a deeper motor of knowledge transformation. It moves library knowledge services from centralized resource storage toward decentralized knowledge becoming.

The driving force behind this movement is the desire of readers and society for flow, connection and creation. The dialectic does not point to a final perfect knowledge order. It points to an ongoing process in which knowledge continually redefines itself through flow, while readers develop new cognitive capacities through experience.

Several limitations should be acknowledged. First, both cases are national libraries whose resources and technological investment exceed those of many public libraries, so the wider applicability of the findings requires further examination. Second, the evaluation of deterritorialization effects relies heavily on quantitative indicators and lacks qualitative tracking of long-term knowledge application and transformation. Third, the paper has only begun to critique technological mediation. Immersive technologies can themselves become new territorializing forces. Knowledge graphs integrate cross-disciplinary relations, but their ontological design may contain classification power; digital watermarking and permission management may transform public knowledge into controllable digital assets (Bowker and Star, 1999; Olson, 2002). Future research should engage more deeply with philosophy of technology.

Further studies can expand the range of cases to libraries of different levels and scales, build fuller evaluation systems for the deterritorialization-reterritorialization process and use interviews and behavioural tracking to assess the long-term influence of immersive knowledge services on readers' cognitive structures. Such work can deepen understanding of the integration of technological empowerment and humanistic value and help libraries build more open, flexible and sustainable knowledge ecosystems.

References

Bekele, M.K., Pierdicca, R., Frontoni, E., Malinverni, E.S. and Gain, J. (2018), "A survey of augmented, virtual, and mixed reality for cultural heritage", *Journal on Computing and Cultural Heritage*, Vol. 11 No. 2, pp. 7:1-7:36, doi: 10.1145/3145534.

Bizer, C., Heath, T. and Berners-Lee, T. (2023), "Linked data - the story so far", in *Linking the World's Information: Essays on Tim Berners-Lee's Invention of the World Wide Web*, 1st ed., Association for Computing Machinery, New York, NY, USA, pp. 115-143.

Bowker, G.C. and Star, S.L. (1999), *Sorting Things Out: Classification and its Consequences*, The MIT Press. doi: 10.7551/mitpress/6352.001.0001.

China Tourism News (2021), "5G panoramic VR Yongle Dadian", 10 December, available at: https://www.ctnews.com.cn/paper/content/202112/10/content_61444.html (accessed 29 April 2026).

Dede, C. (2009), "Immersive interfaces for engagement and learning", *Science*, Vol. 323 No. 5910, pp. 66-69, doi: 10.1126/science.1167311.

Deleuze, G. and Guattari, F. (1983), *Anti-Oedipus: Capitalism and Schizophrenia*, Univ Of Minnesota Press, Minneapolis.

Downloaded from www.emerald.com/jd/article-pdf/doi/10.1108/JD-05-2026-0262/11922832/jd-05-2026-0262en.pdf by Carnegie Mellon University user on 21 October 2026

Deleuze, G. and Guattari, F. (1987), *A Thousand Plateaus: Capitalism and Schizophrenia*, translated by Massumi, B, University of Minnesota Press, Minneapolis, MN.

Deleuze, G. and Guattari, F. (1996), *What Is Philosophy?*, translated by Tomlinson, H. and Burchell, G, Columbia University Press, New York.

Eisenhardt, K.M. (1989), "Building theories from case study research", *Academy of Management Review*, Vol. 14 No. 4, pp. 532-550, doi: 10.5465/amr.1989.4308385.

Faucher, K.X. (2014), "An information meta-state approach to documentation", *Journal of Documentation*, Vol. 70 No. 4, pp. 503-525, doi: 10.1108/JD-09-2012-0118.

Floridi, L., Cowls, J., Beltrametti, M., Chatila, R., Chazerand, P., Dignum, V., Luetge, C., Madelin, R., Pagallo, U., Rossi, F., Schafer, B., Valcke, P. and Vayena, E. (2018), "AI4People — an ethical framework for a good AI society: opportunities, risks, principles, and recommendations", *Minds and Machines*, Vol. 28 No. 4, pp. 689-707, doi: 10.1007/s11023-018-9482-5.

Gerolami, N. (2015), "The library assemblage: creative institutions in an information society", *Journal of Documentation*, Vol. 71 No. 1, pp. 165-174, doi: 10.1108/JD-09-2013-0120.

Holley, R. (2010), "Crowdsourcing: How and why should libraries do it?", *D-lib Magazine*, Vol. 16 Nos 3/4, doi: 10.1045/march2010-holley.

Lemieux, V.L. (2016), "Trusting records: is blockchain technology the answer?", *Records Management Journal*, Vol. 26 No. 2, pp. 111-139, doi: 10.1108/RMJ-12-2015-0042.

Makransky, G. and Petersen, G.B. (2021), "The cognitive affective model of immersive learning (CAMIL): a theoretical research-based model of learning in immersive virtual reality", *Educational Psychology Review*, Vol. 33 No. 3, pp. 937-958, doi: 10.1007/s10648-020-09586-2.

Massis, B. (2015), "Using virtual and augmented reality in the library", *New Library World*, Vol. 116 Nos 11-12, pp. 796-799, doi: 10.1108/NLW-08-2015-0054.

Mayer, R.E. and Moreno, R. (2003), "Nine ways to reduce cognitive load in multimedia learning", *Educational Psychologist*, Vol. 38 No. 1, pp. 43-52, doi: 10.1207/S15326985EP3801_6.

Mittelstadt, B.D., Allo, P., Taddeo, M., Wachter, S. and Floridi, L. (2016), "The ethics of algorithms: mapping the debate', Big Data & Society", *SAGE Publications Ltd*, Vol. 3 No. 2, doi: 10.1177/2053951716679679.

National Center for Preservation and Conservation of Ancient Books (2025), "Continuing the cultural lineage for future generations: ancient-book protection work since the 18th National Congress of the Communist Party of China", available at: https://www.nlc.cn/pcab/xctg/bd/20250327_2644155.shtml (accessed 29 April 2026).

National Library of China (2023), "Yongle Dadian high-definition image database and national catalogue of rare ancient books knowledge base officially released", 13 February, available at: https://www.nlc.cn/web/dsb_zx/gtxw/20230215_2568195.shtml (accessed 29 April 2026).

Nissenbaum, H. (2010), *Privacy in Context: Technology, Policy, and the Integrity of Social Life*, Stanford Law Books, Stanford, CA.

North East Museums (2022), "Lindisfarne gospels exhibition attracts over 50,000 visitors to the Laing art gallery", 13 December, available at: https://www.northeastmuseums.org.uk/news/lindisfarne-gospels-exhibition-attracts-over-50-000-visitors-to-the-laing-art-gallery (accessed 29 April 2026).

Olson, H.A. (2002), *Power to Name: Locating the Limits of Subject Representation in Libraries*, Kluwer Academic, Dordrecht [The Netherlands].

Oomen, J. and Aroyo, L. (2011), "Crowdsourcing in the cultural heritage domain: opportunities and challenges", *Proceedings of the 5th International Conference on Communities and Technologies*, New York, NY, USA, Association for Computing Machinery, pp. 138-149. doi: 10.1145/2103354.2103373.

Piccialli, F. and Chianese, A. (2017), "Cultural heritage and new technologies: trends and challenges", *Personal and Ubiquitous Computing*, Vol. 21 No. 2, pp. 187-189, doi: 10.1007/s00779-016-0984-y.

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Pope, H. (2018), "Chapter 2. Incorporating virtual and augmented reality in libraries", *Library Technology Reports*, Vol. 54 No. 6, pp. 8-11.

Research Center for Digital Humanities of PKU (2023), "Yongle Dadian high-definition image database system", available at: https://pkudh.org/project/yongledadian/ (accessed 29 April 2026).

Ridge, M. (2014), *Crowdsourcing Our Cultural Heritage*, Routledge, Farnham, Surrey GB.

Robinson, L. and Maguire, M. (2010), "The rhizome and the tree: changing metaphors for information organisation", *Journal of Documentation*, Vol. 66 No. 4, pp. 604-613, doi: 10.1108/00220411011052975.

Sweller, J. (1988), "Cognitive load during problem solving: effects on learning", *Cognitive Science*, Vol. 12 No. 2, pp. 257-285, doi: 10.1016/0364-0213(88)90023-7.

Walsham, G. (1995), "Interpretive case studies in IS research: nature and method", *European Journal of Information Systems*, Vol. 4 No. 2, pp. 74-81, doi: 10.1057/ejis.1995.9.

Yin, R.K. (2018), *Case Study Research and Applications: Design and Methods*, Sage Publications, Los Angeles.

Zeng, M.L. (2019), "Semantic enrichment for enhancing LAM data and supporting digital humanities", *Review article*, Vol. 28 No. 1, doi: 10.3145/epi.2019.ene.03.

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